Free fall

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Describe and calculate ideal free fall near Earth.

SUCCESS CRITERIA - I can...

1. State that you are ignoring air resistance.
2. Use equal velocity changes in equal times.
3. Separate acceleration from mass and current velocity.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “free fall” and “initial” to explain your answer. Sentence starters are optional.

free fall: motion with gravity as the only force

initial: at the beginning of the interval; subscript i

acceleration: rate of change of velocity

01 Warm-up: bring back an earlier idea

A simplified calculation predicts position (**-40**, **60**) km with bounds ±**4 km** in x and ±**6 km** in y. **Write a short briefing giving** both intervals and two reasons it is not a real asteroid collision forecast.

Sentence starter: I predict … . I used … to make this prediction because … .

Marzano check - circle: ① ② ③

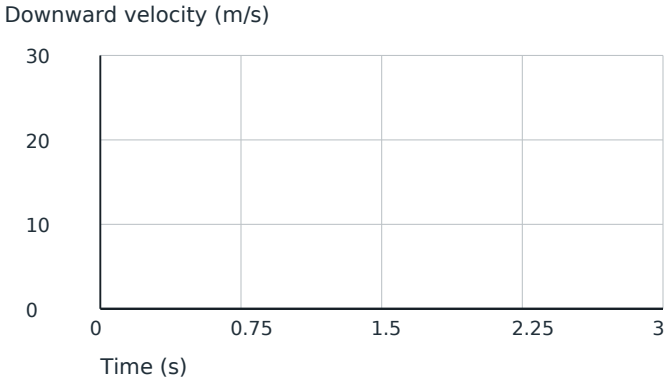
02 Worked example: explain the prediction

Reference: $v_f = v_i + g\,\Delta t$; $\Delta y = v_i\,\Delta t + \frac{1}{2}g(\Delta t)^2$. Downward positive for these equations; g = **10 m/s²**. Ideal free fall: gravity is the only force.

Worked reasoning: A ball dropped from rest has downward speeds **0**, **10**, **20 m/s** at **0**, **1**, **2 s** in ideal free fall with g=**10 m/s²**. Equal times give equal velocity changes.

Your turn: 1. **Sketch its velocity-time graph** with downward positive.
2. **Explain why the positions** are not equally spaced.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two balls of different mass are released together in a vacuum from the same height. **Compare acceleration** and arrival time.

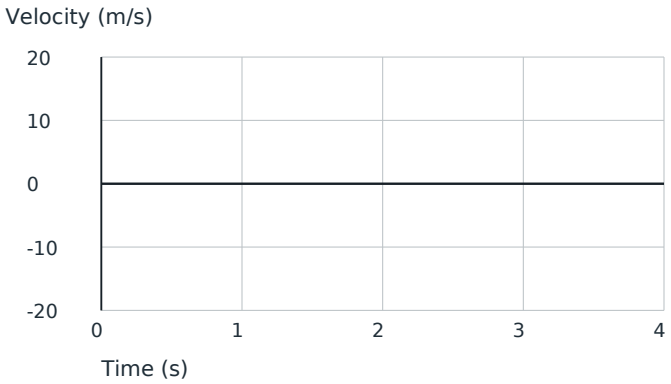
Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

04 Apply the physics

A ball is thrown straight upward at **20 m/s**. Take upward as positive; use g = **10 m/s²** and ignore air resistance. **Complete its velocity-time graph**. **When does** it reach the top, and what is its velocity at **3 s**?

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

At the highest point of a tossed ball, a classmate says gravity stops because speed is zero. Correct the statement.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

An a-versus-F_{net} graph has slope **2** (m/s²)/N. **Find mass** and acceleration at **3 N**. **Explain why a driving-force plot** can have an offset.
Use the table to keep the controlled and changed quantities distinct.

Word help: mass = a measure of how strongly an object resists acceleration

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Trial	Mass	Net force	Acceleration
A	____	____	____
B	____	____	____

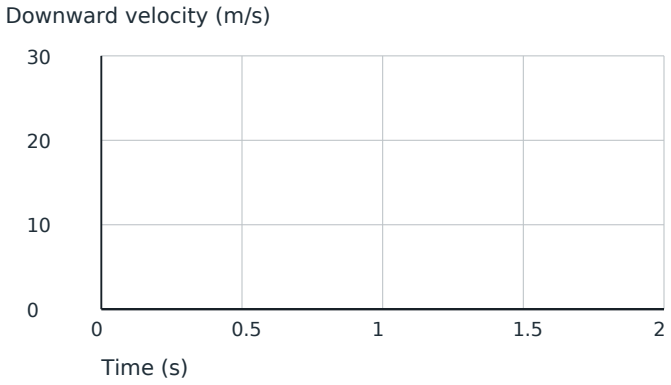
Marzano check - circle: ① ② ③

07 Exit check

A ball starts downward at **5 m/s**. It falls for **2 s** with g = **10 m/s²** and negligible air resistance.
1. **Draw its velocity-time graph**.
2. Use slope and area to **find its final velocity** and downward displacement.

Word help: negligible = small enough to leave out of this model

Sentence starter: I found … [unit]. This value means … in this situation.

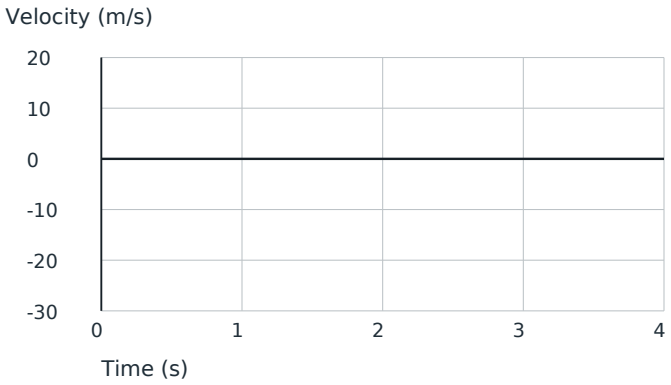


Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A ball is thrown up at **15 m/s** from a **20 m** platform.
1. Use g=**10 m/s²**.
2. **Find its height** at **2 s** and its time to reach the ground.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③